

THOMAS C. MARTINEZ

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RESEARCH INTERESTS

Algebraic combinatorics; total positivity and positroid varieties; Coxeter combinatorics

EDUCATION

University of California, Los Angeles

PhD, Mathematics (expected 2027) · Advisor: Pavel Galashin

2022–present

Los Angeles, CA

Harvey Mudd College

BS, Mathematics · High Distinction and Honors in Mathematics

Thesis: *The Slice Rank Polynomial Method* · Advisor: Mohamed Omar

2017–2021

Claremont, CA

PUBLICATIONS AND PREPRINTS

Preprints

[P1] *The Combinatorics of Affine Deodhar Diagrams*. 2026. arXiv:2607.15672.

[P2] (with B. B. Connelly*, E. Ito*, O. Shevchenko, and K. Yang*) *Unimodality of q -Fibonomial Coefficients for Small Cases*. 2026. arXiv:2605.12822.

Manuscripts in Preparation

[W1] *Parking on Positroids*. Manuscript in preparation.

[W2] *Slice Links and Tori*. (with M. J. Tyler) Manuscript in preparation.

Publications

- [1] (with T. Jaklitsch, S. J. Miller, and S. Mukherjee) *Connections of Class Numbers to the Group Structure of Generalized Pythagorean Triples*. *Rocky Mountain Journal of Mathematics* **55**(2), 369–380, 2025.
- [2] (with T. Blum, C. Choi, A. Hoey, J. Iskander, and K. Lakein) *On Class Numbers, Torsion Subgroups, and Quadratic Twists of Elliptic Curves*. *Transactions of the American Mathematical Society* **375**(1), 351–368, 2022.
- [3] (with H. V. Chu, D. King, N. Luntzlara, S. J. Miller, L. Shao, C. Sun, and V. Xu) *Generalizing the Distribution of Missing Sums in Sumsets*. *Journal of Number Theory* **239**, 402–444, 2022.
- [4] (with S. J. Miller, C. Mizgerd, J. Murphy, and C. Sun) *Generalizing Zeckendorf’s Theorem to Homogeneous Linear Recurrences, II*. *Fibonacci Quarterly* **60**(5), 220–230, 2022.
- [5] (with S. J. Miller, C. Mizgerd, and C. Sun) *Generalizing Zeckendorf’s Theorem to Homogeneous Linear Recurrences, I*. *Fibonacci Quarterly* **60**(5), 231–254, 2022.
- [6] (with F. Beretta, J. Dimino, W. Fang, S. J. Miller, and D. Stoll) *On Benford’s Law and the Coefficients of the Riemann Mapping Function for the Exterior of the Mandelbrot Set*. *Fractal and Fractional* **6**(10), 534–552, 2022.
- [7] (with A. T. Benjamin and J. Lentfer) *Counting on Euler and Bernoulli Number Identities*. *Fibonacci Quarterly* **58**(5), 30–33, 2020.

* Undergraduate collaborator

SELECTED TALKS AND PRESENTATIONS

CMND 2026: Algebraic Combinatorics and Applications

Lightning talk: *Affine Patches of Open Positroid Varieties*

July 2026

Notre Dame, IN

Binghamton University Combinatorics Seminar

The Combinatorics of Affine Deodhar Diagrams

March 2026

Binghamton, NY

Cornell Discrete Geometry and Combinatorics Seminar

The Combinatorics of Affine Deodhar Diagrams

February 2026

Ithaca, NY

CanADAM 2025: Algebraic Combinatorics Minisymposium

Invited minisymposium talk: *Affine Deodhar Diagrams and Rational Dyck Paths*

May 2025

Ottawa, ON

LatMath 2025	March 2025
Poster: <i>Affine Deodhar Diagrams and Rational Dyck Paths</i> (Outstanding Poster Award)	Los Angeles, CA
UCLA Combinatorics Learning Seminar	June 2024
<i>The (Strong and Weak) Bruhat Order</i>	Los Angeles, CA
UCLA Combinatorics Research Seminar	April 2024
<i>Complex Weighted Tilings and Kasteleyn's Theorem</i>	Los Angeles, CA
UCLA Combinatorics Learning Seminar	April 2024
<i>Introduction to Root Systems and Reflection Groups</i>	Los Angeles, CA
UCLA Combinatorics Research Seminar	February 2024
<i>Deograms and Dyck Paths</i>	Los Angeles, CA
UCLA Combinatorics Research Seminar	May 2023
<i>The Slice Rank Polynomial Method</i>	Los Angeles, CA
19th International Fibonacci Conference	July 2020
<i>Generalizing Zeckendorf's Theorem to Homogeneous Linear Recurrences</i>	Virtual
University of Virginia REU Symposium	July 2020
<i>On Class Numbers, Torsion Subgroups, and Quadratic Twists of Elliptic Curves</i>	Virtual
Young Mathematicians Conference	August 2019
Poster: <i>Generalizing the Distribution of Missing Sums in Sumsets</i>	Columbus, OH
UMass Amherst REU Conference	July 2019
<i>Generalizing the Distribution of Missing Sums in Sumsets</i>	Amherst, MA

HONORS/AWARDS

LatMath Outstanding Poster Award	2025
Raymond Redheffer Prize	2025
<i>UCLA departmental award for excellence in calculus teaching.</i>	
Greever Research Prize – Honorable Mention	2021
<i>Harvey Mudd departmental award for original mathematical research.</i>	
Alvin White Prize	2020
<i>For contributions to the Harvey Mudd mathematics community.</i>	

TEACHING

Teaching Assistantships	University of California, Los Angeles
• Math 32A: Multivariable Differential Calculus [x3]	2023–2025
• Math 61: Introduction to Discrete Structures [x7]	2022–2025
• Math 184: Enumerative Combinatorics	Spring 2025
• Math 180: Graph Theory	Fall 2024
• Math 115A: Linear Algebra	Winter 2024
• Math 31B: Integration and Infinite Series	Winter 2024
Teaching Assistantships	Cornell University
• Math 2930: Differential Equations for Engineers	Spring 2026
Tutoring	Harvey Mudd College
• <i>Academic Excellence</i> Peer Tutor	Fall 2019 - Spring 2021
Proof-based Calculus, Linear Algebra, Differential Equations, Multivariable Calculus, Discrete Mathematics	

CONFERENCE, SEMINAR, AND COMMITTEE ORGANIZATION

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| • UCLA Directed Reading Program Committee | September 2025–present |
| • 21st International Fibonacci Conference (Harvey Mudd College – Claremont, CA) | July 2024 |
| • UCLA Combinatorics Learning Seminar (Coxeter Groups) | Spring 2024 |
| • UCLA Combinatorics Research Seminar | Spring 2024 |
| • UCLA Combinatorics Research Seminar | Winter 2024 |
| • 20th International Fibonacci Conference (University of Sarajevo – Sarajevo, Bosnia & Herzegovina) | July 2022 |
| • 19th International Fibonacci Conference (Virtual, via Zoom) | July 2020 |

LEADERSHIP / OUTREACH

UCLA Directed Reading Program	University of California, Los Angeles
<i>Mentor for semester-long undergraduate reading projects</i>	September 2023–present

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| • Brendan Connelly, Ezekiel Ito, Kacey Yang (with Olha Shevchenko): q -Fibonomial coefficients | Spring 2026 |
| • Arjun Mittha: Combinatorial optimization | Spring 2025 |
| • Darsh Verma: The Art of Combinatorial Proof | Spring 2025 |
| • Thomas Slavonia: Schubert polynomials and pipe dreams | Spring 2024 |
| • Thomas Slavonia: Representation theory of the symmetric group | Winter 2024 |
| • Anika Misra: Graph theory | Winter 2024 |
| • Akash Kote: The probabilistic method in combinatorics | Fall 2023 |

UCLA Learning Assistant Program	University of California, Los Angeles
<i>Math Learning Assistant Coordinator</i>	September 2024–September 2025

- Facilitated discussions on mathematical pedagogy and resolved course-specific content questions.
- Oversaw the Learning Assistant Program, promoting collaborative learning in weekly discussion sections.

The Motivational Edge	Allapattah, FL
<i>Volunteer Tutor</i>	September 2021–March 2022

- Provided weekly tutoring in mathematics, English, and science to elementary and middle school students.
- Collaborated with directors to develop personalized learning plans based on student progress and needs.

City Year Miami	Miami Jackson Senior High School
<i>Team Leader and Math AmeriCorps Member</i>	August 2021–June 2022

- Led and managed a team of four AmeriCorps members, delivering academic tutoring, behavioral interventions, and school-wide initiatives to over 200 ninth- and tenth-grade students.
- Provided targeted support in Algebra 1 to 71 students, focusing on attendance, behavior, and academic improvement.
- Designed and implemented weekly interventions for 11 students to address critical academic and behavioral challenges.

Number Theory & Benford's Law Research Groups	Polymath Jr. Program (Virtual)
<i>Assistant Director</i>	June 2021–August 2021

- Co-mentored 40 students alongside Professor Steven J. Miller and other researchers, guiding them through advanced research topics.
- Organized and facilitated a weekly seminar series for students to present research progress and receive feedback.
- Contributed to the production of several publications and conference presentations by program participants.

Associated Students of Harvey Mudd College	Harvey Mudd College
<i>West Dorm Co-President</i>	April 2018–April 2019

- Represented West Dorm in Student Government, advocating for residents' interests and concerns.
- Co-led monthly dorm meetings, coordinating with fellow Co-Presidents to address community needs and enhance dorm life.

LANGUAGES

Programming/Markup:

- SageMath · Python · Java · C++ · $\text{\LaTeX}$

Written/Spoken:

- English (native) · Portuguese (native) · Spanish (fluent) · French (basic)

Last updated: August 25, 2026